1.2 Motion - Mark scheme

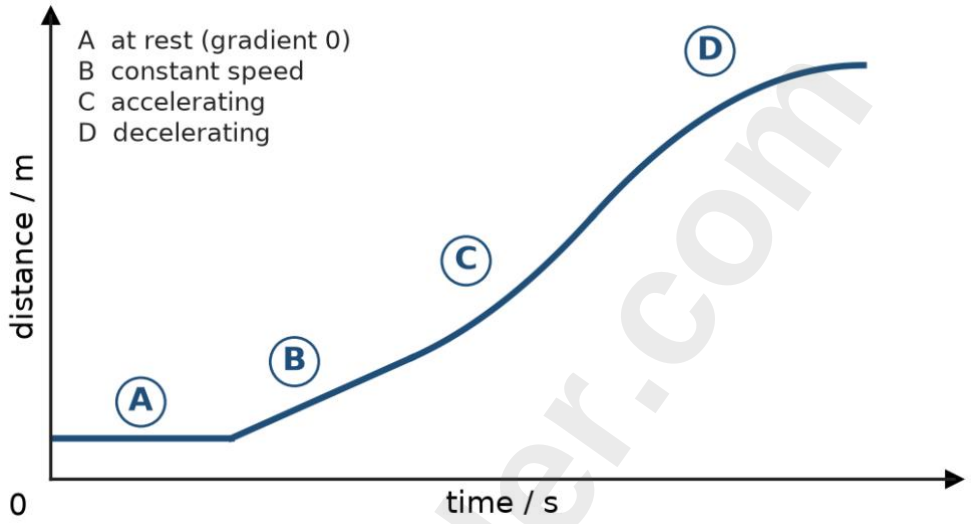
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Tier: Core and Extended

For teachers. Contains the answers to Sections 6 and 7 and model versions of the Section 4 diagrams.

Model diagrams (Section 4)

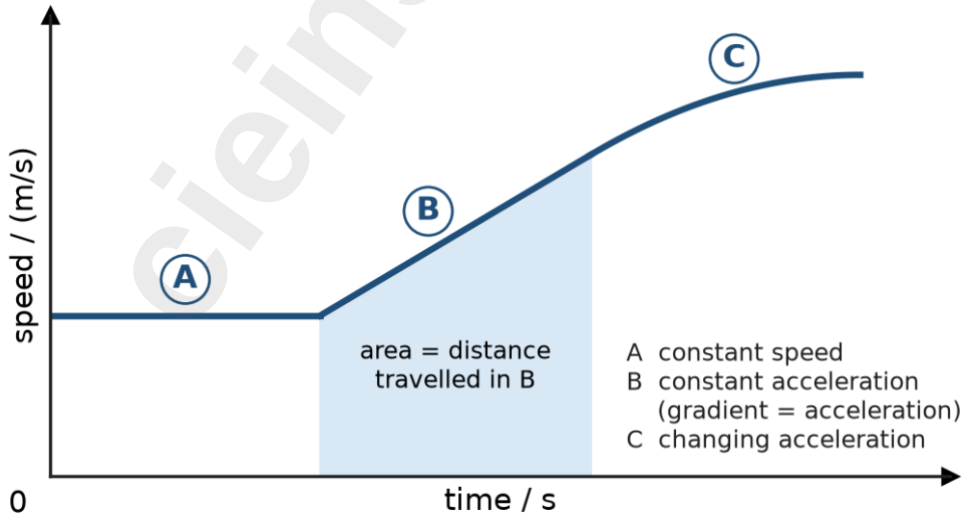
4.1 Distance–time graph Core



Gradient of a distance–time graph = speed. The line never slopes downwards for a journey in one direction.

Credit for: A is a horizontal line (gradient zero); B is a straight line sloping upwards; C curves upwards (gradient increasing); D curves over (gradient decreasing); Stages labelled A, B, C, D.

4.2 Speed–time graph Core

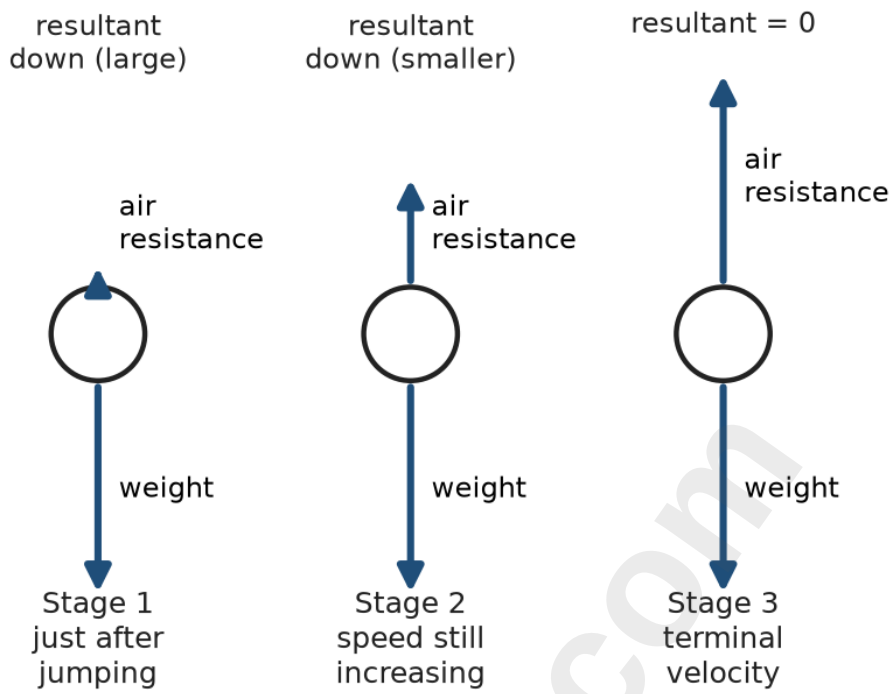


Gradient = acceleration; area under the line = distance travelled. Identifying changing acceleration (stage C) is Extended.

Credit for: A is a horizontal line above the time axis; B is a straight line sloping upwards; Extended: C is a curve (gradient changing); Area under B shaded and labelled 'distance'.

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4.3 Forces on a falling object Extended



Weight is constant. Air resistance increases with speed, so the resultant (downward) force and the acceleration decrease until air resistance = weight at terminal velocity.

Credit for: Weight arrow pointing down, the same length at every stage; Air resistance arrow pointing up, getting longer from stage 1 to stage 3; Stage 3: air resistance arrow equal in length to weight; Both forces labelled.

Multiple-choice answers (Section 6)

Each wrong option is a specific mistake. Where an examiner report records candidates making it, the series and paper are given in brackets. Use the third column to diagnose, not just to mark.

#	Answer	Why each wrong option is wrong
1	**B**	**A** This is speed. It leaves out the direction. **C** This is acceleration. **D** Treats velocity as a change in speed. (June 2024, Paper 12)
2	**B**	**A** Uses the uphill stage only (3000 m ÷ 600 s). **C** Averages the two speeds, (5.0 + 10) ÷ 2, instead of total distance ÷ total time. (June 2021, Paper 23) **D** Uses the downhill stage only (3000 m ÷ 300 s).
3	**D**	**A** Picks the graph that looks the same as the speed–time graph. A horizontal distance–time line means the car is at rest. (June 2023, Paper 12) **B** Gradient increasing: the car would be accelerating. **C** Distance from the start decreasing: the car would be coming back.
4	**C**	**A** Works out the gradient (the acceleration, 20 ÷ 8.0) instead of the area. **B** Reads the final speed as if it were a distance. **D** Multiplies the final speed by the time (20 × 8.0): a rectangle, not the triangle under the line. (Nov 2022, Paper 13 and June 2023, Paper 12)

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#	Answer	Why each wrong option is wrong
5	**A**	**B** Thinks the acceleration of free fall increases as the object falls. (March 2022, Paper 12 and Nov 2024, Paper 21) **C** Thinks the acceleration of free fall decreases as the object falls. (Nov 2024, Paper 11) **D** Terminal velocity needs air resistance, and the question says it is negligible.
6	**C**	**A** Divides the top speed by the whole time on the graph (8.0 ÷ 9.0). **B** Includes the first second, when the athlete is still at rest (8.0 ÷ 5.0). The acceleration only lasts from t = 1.0 s to t = 5.0 s. **D** Works out the area under the sloping line (½ × 4.0 × 8.0) instead of the gradient. (June 2024, Paper 22)
7	**D**	**A** Multiplies the change in velocity by the time (10 × 0.20) instead of dividing. **B** Subtracts the speeds (6.0 − 4.0 = 2.0 m/s) and ignores the change of direction. The change in velocity is 4.0 − (−6.0) = 10 m/s. (March 2021, Paper 22 and June 2021, Paper 21) **C** Divides the average of the two speeds by the time (5.0 ÷ 0.20).
8	**A**	**B** Thinks the speed keeps increasing after terminal velocity is reached. (June 2022, Paper 23) **C** True only just after a parachute opens, when the skydiver slows down. **D** Confuses a zero resultant force with no forces. Weight and air resistance both act and are equal.

Mark scheme (Section 7)

Question 1 [1]

g = 9.8 m/s² **[1]**

Not 10 m/s². The question asks for an acceleration, so the unit is m/s²; 9.8 N/kg is gravitational field strength (1.3 Mass and weight).

Question 2 [2]

Speed is distance travelled per unit time / has magnitude only (scalar) **[1]**

Velocity is speed in a given direction (vector) **[1]**

Question 3 [3]

Converts units: 150 km = 150 000 m and 2.5 h = 9000 s **[1]**

average speed = 150 000 / 9000 **[1]**

= 16.7 m/s (accept 17 m/s) **[1]**

Question 4 [5]

(a) First section (rectangle): 12 × 6.0 = 72 m **[1]**

Second section (trapezium): ½ × (12 + 30) × 6.0 = 126 m **[1]**

Total distance = 72 + 126 = 198 m **[1]**

(b) a = Δv / Δt = (30 − 12) / (12 − 6.0) = 18 / 6.0 **[1]**

a = 3.0 m/s² **[1]**

Question 5 [2]

A curve whose gradient decreases (becomes less steep) **[1]**

The line does not have to become horizontal to gain this mark.

The gradient is the speed, so the speed is decreasing **[1]**

Question 6 [3] Extended

Acceleration is the change in velocity per unit time (or rate of change of velocity) **[1]**

Do not credit a definition that mixes 'rate of change' with 'per unit time'.

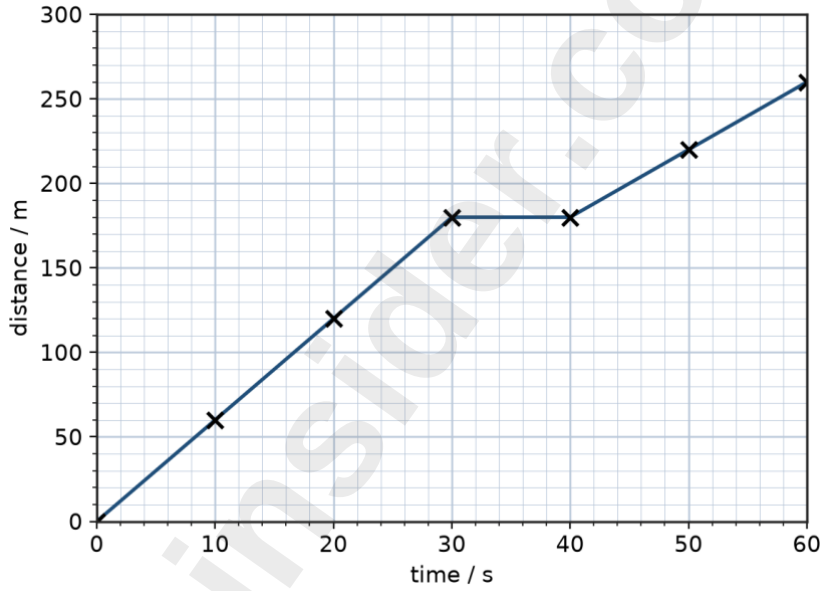
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a = (22.0 − 4.0) / 3.0 = 18.0 / 3.0 **[1]**
a = 6.0 m/s² **[1]**

Question 7 [8] Extended

(a) Speed increases at a constant rate / constant acceleration (of g = 9.8 m/s²) **[1]**
(b) As the speed increases, the air resistance increases **[1]**
The resultant (downward) force decreases **[1]**
So the acceleration decreases **[1]**
Terminal velocity when air resistance = weight / resultant force = 0 (so acceleration = 0) **[1]**
Weight stays constant: do not credit 'gravity decreases' or 'weight decreases'.
(c) Any three of the four points below, max 3.
Air resistance increases (larger surface area) and becomes greater than the weight **[1]**
Resultant force is upwards, so the skydiver decelerates / slows down **[1]**
As the speed decreases, the air resistance decreases (until it equals the weight again) **[1]**
New terminal velocity, lower than before **[1]**
Do not credit 'the skydiver moves upwards': the velocity stays downwards throughout.

Question 8 [8]



(a) All seven points plotted to within half a small square **[2]**
One point wrong: 1 mark.
Straight lines joining the points with a ruler **[1]**
(b) Stationary / at rest **[1]**
(c) speed = gradient = 180 / 30 **[1]**
= 6.0 m/s **[1]**
(d) average speed = 260 / 60 **[1]**
= 4.3 m/s **[1]**

Question 9 [3] Extended

a = Δv / Δt = (0 − 8.0) / 2.5 **[1]**
a = −3.2 m/s² **[1]**
Negative sign: the acceleration is opposite to the motion / the cyclist is decelerating (slowing down) **[1]**

Question 10 [6] Extended

(a) P **[1]**
(b) Speed increasing **[1]**

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Acceleration decreasing / changing (gradient decreasing) **[1]**

(c) a = gradient = 12 / 4.0 **[1]**

= 3.0 m/s² **[1]**

(d) 0 / zero (constant speed) **[1]**

Syllabus coverage

Where each checklist statement is practised in this booklet.

#	Statement	Practised in
1	Define speed as distance travelled per unit time; recall and use the equation v = s/t	WE1, Q3, Q8(c), MCQ2
2	Define velocity as speed in a given direction	Q2, MCQ1
3	Recall and use the equation: average speed = total distance travelled / total time taken	Q3, Q8(d), MCQ2
4	Sketch, plot and interpret distance–time and speed–time graphs	Diagrams 4.1, 4.2, Q8(a), Q10, MCQ3
5	Determine, qualitatively, from given data or the shape of a distance–time graph or speed–time graph when an object is: (a) at rest (b) moving with constant speed (c) accelerating (d) decelerating	Diagram 4.1, Q5, Q8(b), MCQ3
6	Calculate speed from the gradient of a straight-line section of a distance–time graph	WE1, Q8(c)
7	Calculate the area under a speed–time graph to determine the distance travelled for motion with constant speed or constant acceleration	WE2, Q4(a), MCQ4
8	State that the acceleration of free fall g for an object near to the surface of the Earth is approximately constant and is approximately 9.8 m/s²	Q1, MCQ5
9	Define acceleration as change in velocity per unit time; recall and use the equation a = Δv/Δt	Q6, MCQ7
10	Determine from given data or the shape of a speed–time graph when an object is moving with: (a) constant acceleration (b) changing acceleration	Diagram 4.2, Q10(a), (b)
11	Calculate acceleration from the gradient of a speed–time graph	WE3, Q4(b), Q10(c), MCQ6
12	Know that a deceleration is a negative acceleration and use this in calculations	Q9, MCQ7
13	Describe the motion of objects falling in a uniform gravitational field with and without air/liquid resistance, including reference to terminal velocity	Diagram 4.3, Q7(a), (b), (c), MCQ8

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